

# torch.argwhere#

<https://pytorch.org/docs/stable/generated/torch.argwhere.html#torch.argwhere>

## Description:

Returns a tensor containing the indices of all non-zero elements of input. Each row in the result contains the indices of a non-zero element in input. The result is sorted lexicographically, with the last index changing the fastest (C-style). If input has  $n$  dimensions, then the resulting indices tensor out is of size  $(z \times n)$ , where  $z$  is the total number of non-zero elements in the input tensor. This function is similar to NumPy's `argwhere`. When input is on CUDA, this function causes host-device synchronization.

## Function Signature

---

`torch.argwhere(input) → Tensor#`

Parameters

## Code Examples

---

```
>>> t = torch.tensor([1, 0, 1])
>>> torch.argmax(t)
tensor([2])
>>> t = torch.tensor([[1, 0, 1], [0, 1, 1]])
>>> torch.argmax(t)
tensor([[0, 0],
        [0, 2],
        [1, 1],
        [1, 2]])
```

## Notes & Warnings

---

**NOTE:** This function is similar to NumPy's `argwhere`. When input is on CUDA, this function causes host-device synchronization.

# Detailed Documentation

---

## Documentation

Returns a tensor containing the indices of all non-zero elements of input. Each row in the result contains the indices of a non-zero element in input. The result is sorted lexicographically, with the last index changing the fastest (C-style).

If input has  $n$  dimensions, then the resulting indices tensor out is of size  $(z \times n) \times n$ , where  $z$  is the total number of non-zero elements in the input tensor.

This function is similar to NumPy's `argwhere`.

When input is on CUDA, this function causes host-device synchronization.